

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
Second Semester, Session: 2025 – 2026
Control Systems (ECE/EEE/INSTR F242)

MM: 6

Time: 12 Minutes

Quiz-1 (Solution)

Date 27.01.2026

Note: Write your final answers in the space provided only. Show all the necessary steps in solution.

ID: SOLUTION

Name: SOLUTION

Section:

1. For the translational mechanical system shown in Fig. Q1,
 (a) Write differential equations governing the dynamic behavior of the system.

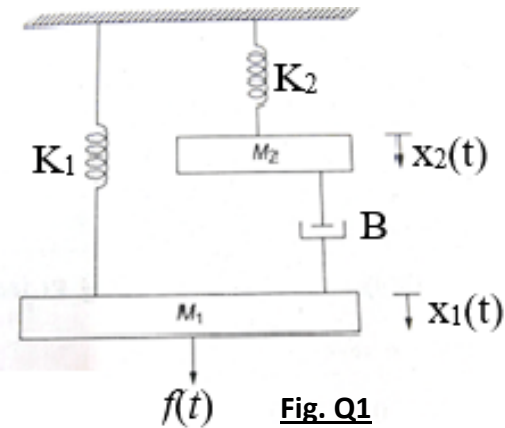
For Mass M_1

$$M_1 \ddot{x}_1 + B(\dot{x}_1 - \dot{x}_2) + K_1 x_1 = f(t)$$

For Mass M_2

$$M_2 \ddot{x}_2 + B(\dot{x}_2 - \dot{x}_1) + K_2 x_2 = 0$$

(1+1 = 2 Marks)



- (b) Find the transfer function, $G(s) = X_2(s)/F(s)$. Given, $M_1 = M_2 = 1 \text{ kg}$, $K_1 = K_2 = 1 \text{ N/m}$, and $B = 1 \text{ N-s/m}$. Use the given numerical values for this part (i.e. Q1(b)) only.

$$(M_1 s^2 + Bs + K_1)X_1(s) - BsX_2(s) = F(s) \Rightarrow (s^2 + s + 1)X_1(s) - sX_2(s) = F(s)$$

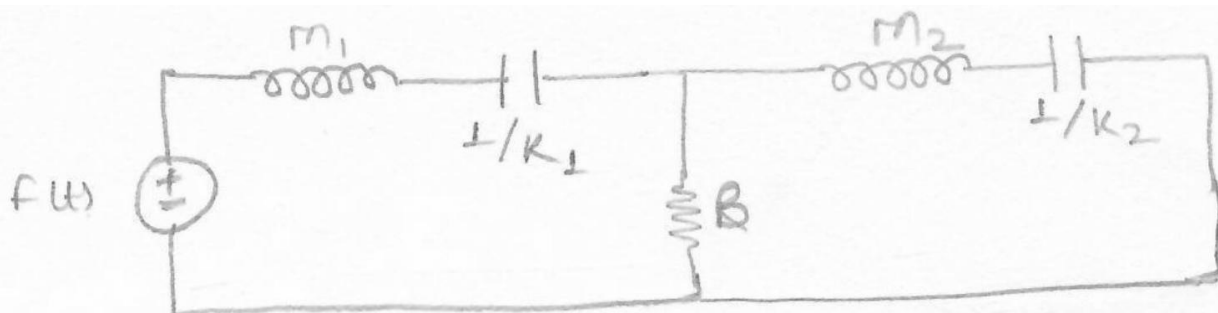
$$-BsX_1(s) + (M_2 s^2 + Bs + K_2)X_2(s) = 0 \Rightarrow -sX_1(s) + (s^2 + s + 1)X_2(s) = 0$$

$$X_2(s) = \frac{\begin{vmatrix} (s^2 + s + 1) & F(s) \\ -s & 0 \end{vmatrix}}{\begin{vmatrix} (s^2 + s + 1) & -s \\ -s & (s^2 + s + 1) \end{vmatrix}} = -\frac{sF(s)}{(s^2 + s + 1)^2 - s^2}$$

$$G(s) = \frac{X_2(s)}{F(s)} = \frac{s}{(s^2 + s + 1)^2 - s^2} = \frac{s}{s^4 + 2s^3 + 2s^2 + 2s + 1}$$

(2 Marks)

- (c) Draw the equivalent electrical analogous system using force-voltage (F-V) analogy.



(2 Marks)